PRACTICAL 13-A

Object Oriented Programming

AIM:

To implement and understand the concepts specified in Practical 13 of the CEUE203 Object Oriented Programming practical list, using Java. The work covers Part A: Generic repository / bounded generics and Part B: Generic Repository<T> using @Id reflection.

CODE SCREENSHOT:

GenericRepository.java

Part A > Practical 13-A > GenericRepository.java > ...

```java
import java.lang.annotation.*;
import java.lang.reflect.*;
import java.util.*;

@Retention(RetentionPolicy.RUNTIME)
@Target(ElementType.FIELD)
@interface Id {
}

class Account {
    @Id
    private final String accountNumber;
    private final String owner;
    private long balance;

    Account(String n, String o, long b) {
        accountNumber = n;
        owner = o;
        balance = b;
    }

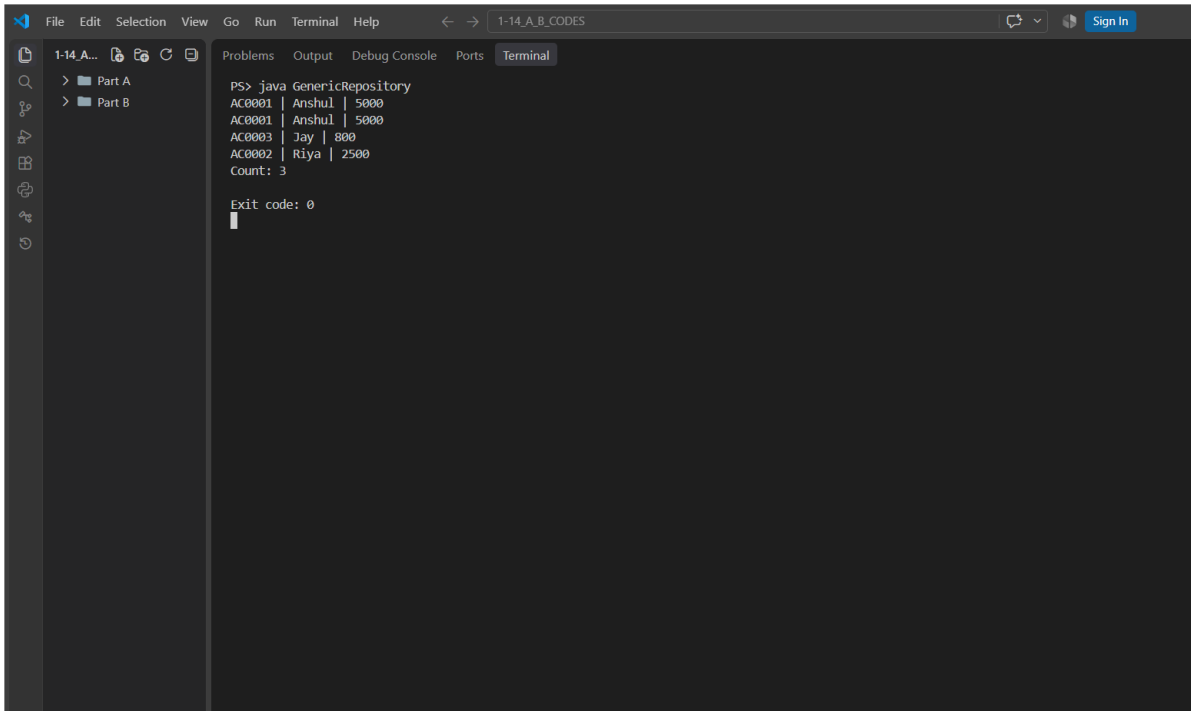
    public String toString() {
        return accountNumber + " | " + owner + " | " + balance;
    }
}

class Repository<T> {
    private final Map<String, T> data = new HashMap<>();

    public T save(T entity) {
        try {
            for (Field f : entity.getClass().getDeclaredFields())
                if (f.isAnnotationPresent(Id.class)) {
                    f.setAccessible(true);
                    data.put(String.valueOf(f.get(entity)), entity);
                    return entity;
                }
            throw new IllegalArgumentException("No @Id field");
        } catch (IllegalAccessException e) {
            throw new RuntimeException(e);
        }
    }

    @SafeVarargs
    public final void saveAll(T... items) {
        for (T item : items)
            save(item);
    }
```

OUTPUT SCREENSHOT:



LEARNING OUTCOMES:

Build reusable type-safe generic components and access entity identifiers using reflection.